

The segment: comparing and measuring

Two ends, one number — and the one property that makes measurement possible.

LESSON 3–4

2 × 40 MIN

MATEMATIKA 7, §2

S8 5

LESSON CLOCK

15'

25'

25'

20'

5'

Comparing	15 min
Measuring	25 min
Adding segments	25 min
The midpoint	20 min
Homework	5 min

LEARNING OBJECTIVES

- Compare two segments by superposition and by measurement.
- Use the additive property $AB = AC + CB$ when C lies between A and B .
- Find the midpoint of a segment and use it.
- Convert between the units of length and solve simple problems.

TEXTBOOK

National	pp. 9–14
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

01

Explanation

Comparing

Two segments are **equal** if one can be laid exactly on the other. In practice they are compared with compasses: open the compasses to AB and see whether the opening reaches from C to D .

RESULT	WRITTEN
the opening fits exactly	$AB = CD$
the opening falls short	$AB < CD$
the opening overshoots	$AB > CD$

COMPARISON COMES BEFORE MEASUREMENT

Two segments can be compared without any ruler at all. Measurement is comparison with one chosen segment — the unit — repeated. That is the whole idea of a length.

Measuring

Choose a **unit segment**. The length of AB is the number of units it contains.

UNIT	SYMBOL	IN METRES
millimetre	mm	0.001
centimetre	cm	0.01
decimetre	dm	0.1
metre	m	1
kilometre	km	1000

Each step up the table is a factor of 10 , except metre to kilometre, which is 1000 .

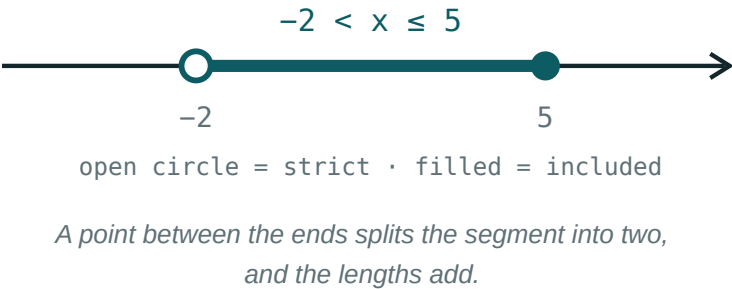
A LENGTH IS A NUMBER AND A UNIT

“The segment is 7” means nothing until the unit is said. Every answer in this chapter carries its unit, and a missing unit costs a mark.

Adding segments

If C lies **between** A and B , then

$$AB = AC + CB$$



AC	CB	AB
3 cm	5 cm	8 cm
4.5 cm	2.5 cm	7 cm
12 mm	$?$	$30\text{ mm} \Rightarrow 18\text{ mm}$

“BETWEEN” IS A CONDITION, NOT A DECORATION

If C is not between A and B the equation is false. With A, B, C in that order, $AC = AB + BC$ instead — so the order of the points must be established from the figure first.

The midpoint

The **midpoint** M of AB is the point between A and B with $AM = MB$. Then

$$AM = MB = \frac{AB}{2}$$

GIVEN	FIND	WORKING
$AB = 14\text{ cm}$	AM	7 cm
$AM = 6.5\text{ cm}$	AB	13 cm
M and N midpoints of AC and CB	MN	$\frac{AB}{2}$

The third row is a small surprise worth checking: whatever the position of C between A and B , the distance between the two midpoints is always half of AB .

A SEGMENT HAS EXACTLY ONE MIDPOINT

Not two, and never none. The construction with compasses — equal arcs from both ends — produces it exactly, and it is the first construction of the course.

02 Worked examples

EXAMPLE 1 C lies between A and B . If $AC = 12\text{ mm}$ and $AB = 30\text{ mm}$, find CB .

① $AB = AC + CB$

② $30 = 12 + CB$

③ $CB = 18\text{ mm}$

④ Check: $12 + 18 = 30\checkmark$

ANSWER

18 mm

EXAMPLE 2 $AB = 14\text{ cm}$ and M is its midpoint. N is the midpoint of AM . Find NB .

① $AM = 7\text{ cm}$

② $AN = 3.5\text{ cm}$

③ $NB = AB - AN = 14 - 3.5$

④ $= 10.5\text{ cm}$

ANSWER

10.5 cm

EXAMPLE 3 Convert 2.4 m to centimetres and 35 mm to metres.

① $1\text{ m} = 100\text{ cm}$

② $2.4 \times 100 = 240\text{ cm}$

③ $1\text{ mm} = 0.001\text{ m}$

④ $35 \times 0.001 = 0.035\text{ m}$

ANSWER

240 cm and 0.035 m

03 Interactive model

Compare two segments drawn on the board with compasses only, then measure both; the class sees that comparison needs no numbers and measurement needs a unit.

Points on a segment

$$x > 2$$



inequality $x > 2$

boundary **2 (open)**

$x = 0$ works? **no**

$x = 6$ works? **yes**

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Segment	Kesma	Отрезок
Length	Uzunlik	Длина
Midpoint	O‘rta nuqta	Середина
Between	Orasida	Между
To superpose	Ustma-ust qo‘ymoq	Наложить
Equal segments	Teng kesmalar	Равные отрезки
Unit of length	Uzunlik birligi	Единица длины
Additive property	Additivlik xossasi	Свойство аддитивности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Two segments are equal when:

they look alike

one lays exactly on the other

they are near

they are drawn together

$AB = AC + CB$ requires:

C anywhere

C between A and B

C outside

nothing

$AM = MB$ makes M :

an end

the midpoint

outside

a ray

$AB = 14\text{ cm}$: $AM =$

1 m equals:

A length must have:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $AC = 3, CB = 5: AB$

ANSWER 8

2. $AC = 12, AB = 30: CB$

ANSWER 18

3. $AB = 14: AM$

ANSWER 7

4. $AM = 6.5: AB$

ANSWER 13

5. 2.4 m in cm

ANSWER 240

6. 35 mm in m

ANSWER 0.035

7. 3 km in m

ANSWER 3000

Medium · the standard question

7 PROBLEMS

1. $AB = 14$, M midpoint, N midpoint of AM : NB

ANSWER 10.5

2. $AC = 4.5$, $CB = 2.5$: AB

ANSWER 7

3. $AB = 25$, $AC = 9$: CB

ANSWER 16

4. 850 cm in m

ANSWER 8.5

5. 0.6 km in cm

ANSWER 60 000

6. A segment of 18 cm divided in $2 : 1$

ANSWER 12 and 6

7. Midpoints of AC and CB with $AB = 20$: MN

ANSWER 10

Hard · stretch and reason

7 PROBLEMS

1. $AB = 24$, C with $AC : CB = 3 : 5$: AC

ANSWER 9

2. Three points A , B , C with $AB = 5$, $BC = 3$: possible AC

ANSWER 8 or 2

3. M is the midpoint of AB and $AM = 3x - 1$, $MB = x + 7$: AB

ANSWER 22

4. A segment is divided into 4 equal parts; the distance between the 1st and 3rd division points, if $AB = 32$

ANSWER 16

5. $AB = 30$ with C between; AC exceeds CB by 6: AC

ANSWER 18

6. A ruler of 30 cm measures a table 7 times exactly: the table

ANSWER 2.1 m

7. Why is the midpoint unique?

ANSWER Two midpoints would give two different halves

07 Homework

Homework — 5 tasks

Every answer carries a unit, and every figure carries its labels.

- C lies between A and B with $AC = 17\text{ mm}$ and $AB = 42\text{ mm}$. Find CB .
- $AB = 18\text{ cm}$ and M is its midpoint. Find AM and MB .
- Convert 4.7 m to millimetres and 2500 cm to kilometres.
- A segment of 45 cm is divided in the ratio $2 : 3$. Find the two parts.

5. Three points lie on a line with $AB = 7$ and $BC = 4$. Give both possible values of AC .